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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-49

AC Type-2 Charging - II

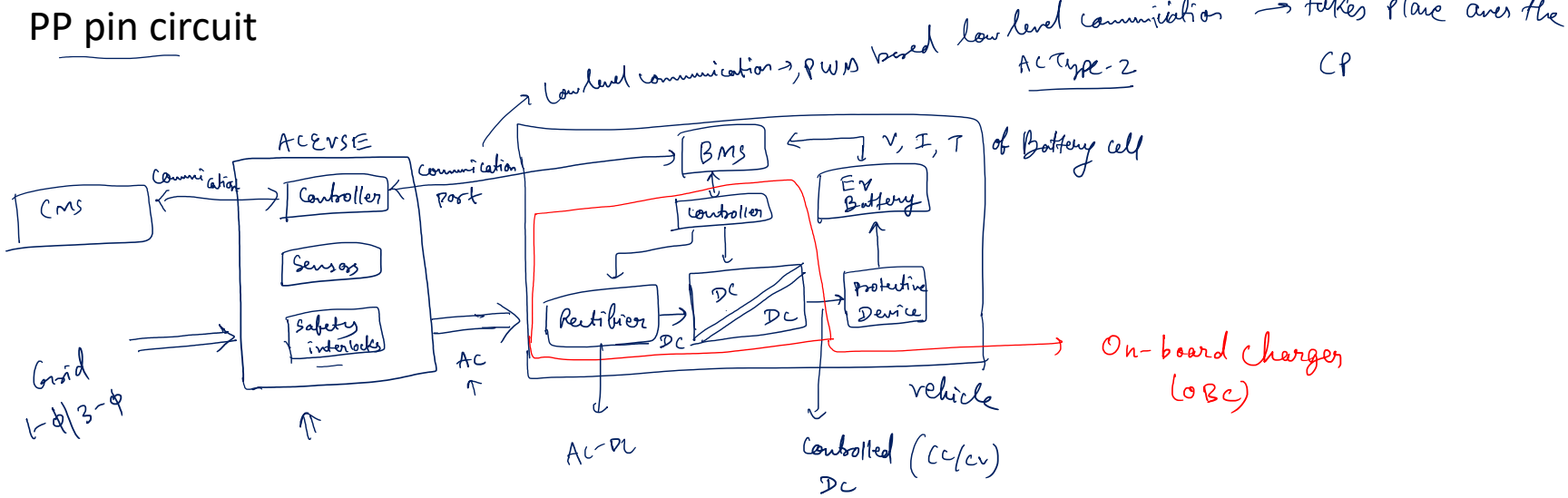
Dr. Apurv Kumar Yadav

Department of Electrical Engineering



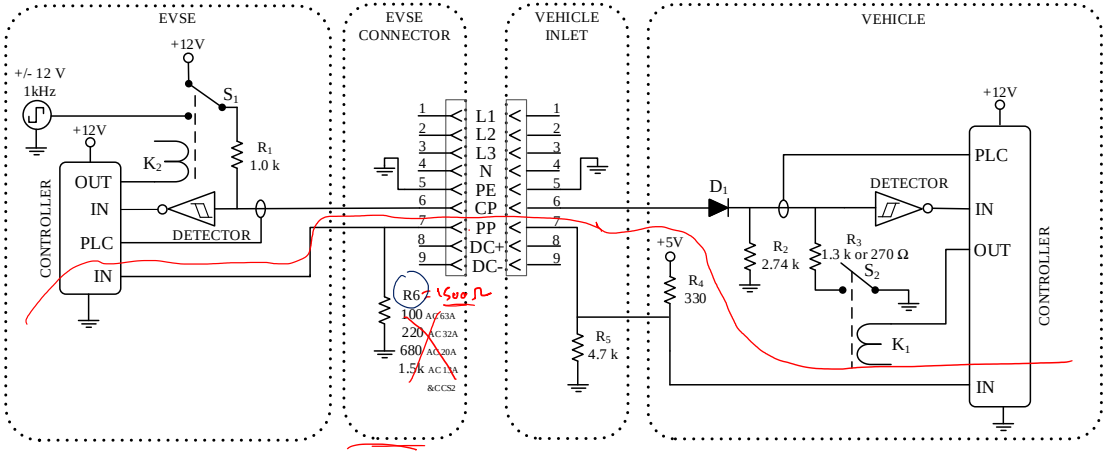
Recap

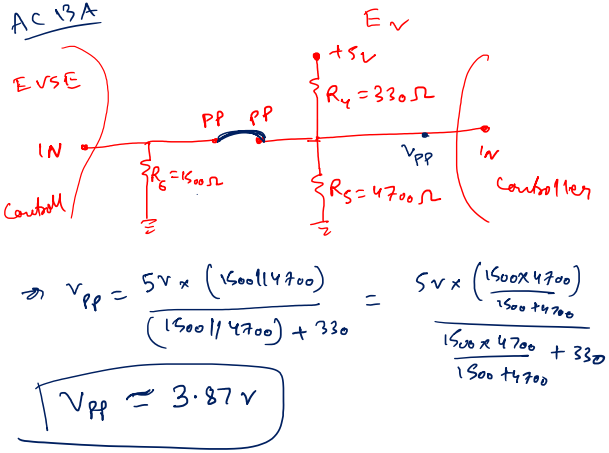
- PP pin circuit



Recap: AC Charging (Type-2 or CCS2): PP Circuit

- PP detected the adaptor (in the EV)





The details are in given in SAEJ1772 standards (in the form of SAE J1772 signalling protocol) and also adopted in IEC 61851

Recap: AC Charging (Type-2 or CCS2): PP Circuit

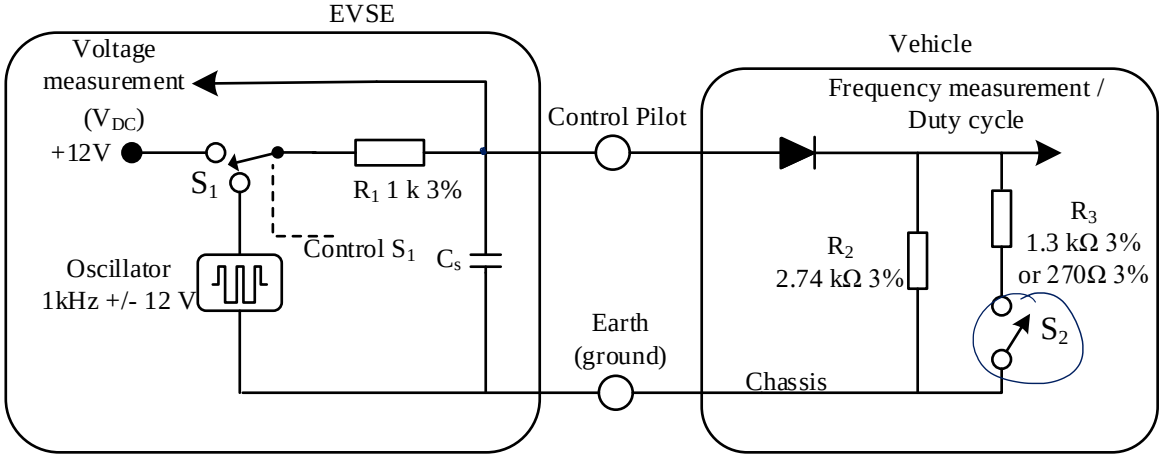
Proximity resistance circuitry tolerances in SAE J1772 and IEC 61851-1

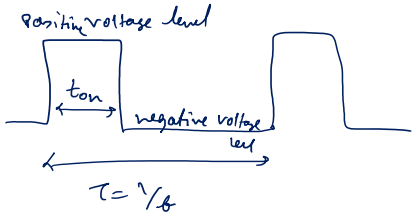
Resistance between PP-PE	Tolerance range	Max. Current	Conductor Size	Rated voltage on PP pin
> 4500 Ω	-	Error condition or disconnected plug	-	> 4.37V
1500 Ω	1100-2460	13A	1.5mm^2	3.87V
680 Ω	400-936	20A	2.5mm^2	3.21V
220 Ω	164-308	32A	6mm^2	1.94V
100 Ω	80-140	63A(3-phase)	16mm^2	1.14V
<60 Ω	-	Error condition	-	<0.76V

Bahrami, Ali. (2020). EV Charging Definitions, Modes, Levels, Communication Protocols and Applied Standards. 10.13140/RG.2.2.15844.53123/11.

AC Charging (Type-2 or CCS2) : CP Circuit

- Over the CP (Control Pilot), a PWM of 1000Hz is generated by the EVSE
- EV measures the duty ratio and frequency
- EV changes the voltage level of PWM pulses by controlling S2
- EVSE measures the peak value of voltage (defined by the EV)



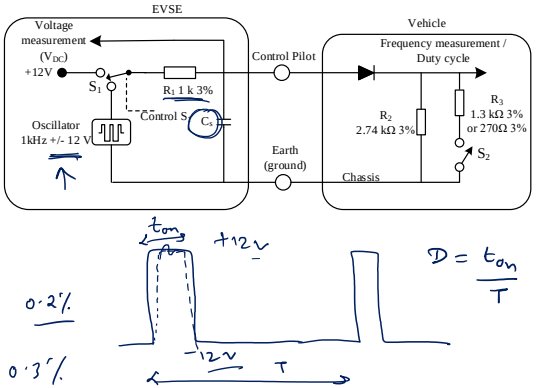


The details are in given in SAEJ1772 standards (in the form of SAE J1772 signalling protocol) and also adopted in IEC 61851

AC Charging (Type-2 or CCS2): CP Circuit Parameters

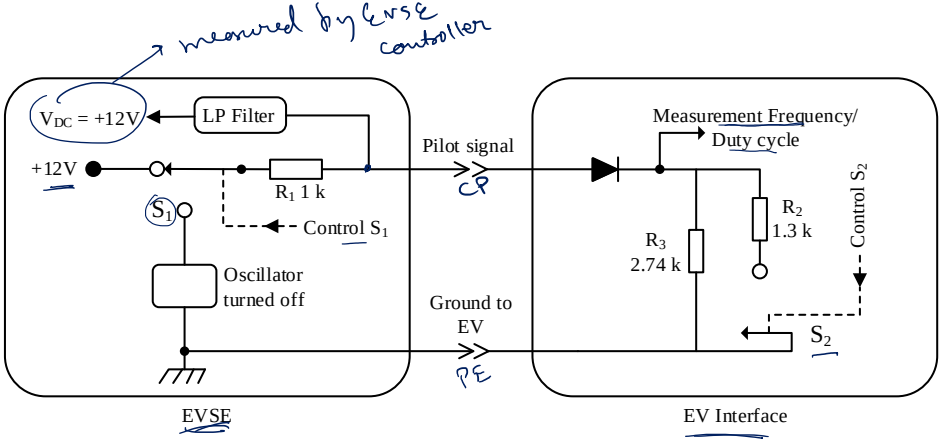
CP Circuit parameters

Parameter	Symbol	Nominal Value	Unit
Positive idle Voltage	U_{och}	12 (+/- 0.6)	V
Negative idle Voltage	U_{ocl}	-12 (+/- 0.6)	V
Frequency	f	1000 (+/- 0.5)	Hz
Maximum value of rising time(10% to 90%)	T_{rg}	2	µs
Maximum value of falling time(10% to 90%)	T_{fg}	2	µs
Minimum value of settling time	T_{sg}	3	µs
Resistance of source	R_1	1000+/-3%	Ω
Recommended capacitance for interference suppression	C_s	300	pF
Maximum value of total line capacity	$C_s + C_c$	3100	pF



Bahrami, Ali. (2020). EV Charging Definitions, Modes, Levels, Communication Protocols and Applied Standards. 10.13140/RG.2.2.15844.53123/11.

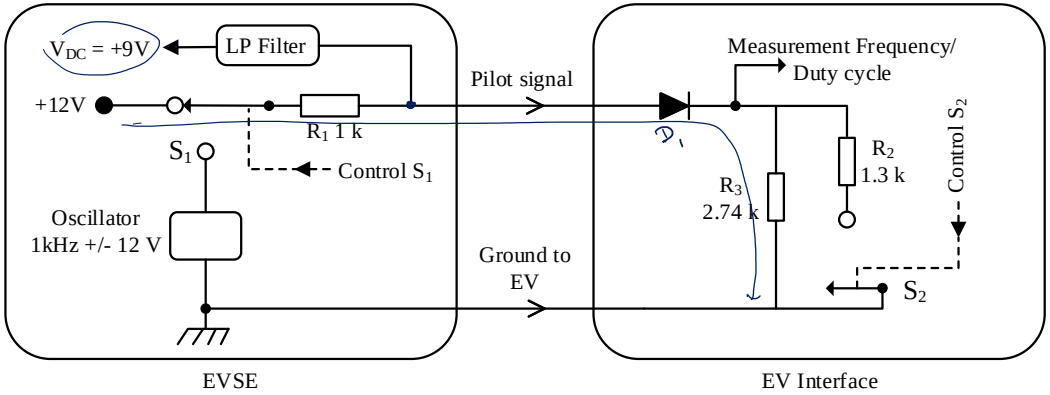
AC Charging (Type-2 or CCS2): Charging Sequence

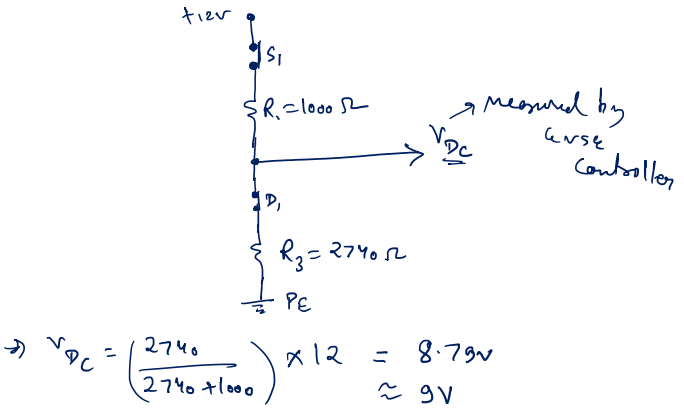


- State-A
- V_{DC} = +12V
- EV is not connected
- PWM is deactivated

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
2. : M. Parchomiuk, et.al, "An Overview of Electric Vehicles Fast Charging Infrastructure," *2019 Progress in Applied Electrical Engineering (PAEE)*, Koscielisko, Poland, 2019, pp. 1-5, doi: 10.1109/PAEE.2019.8788983

AC Charging (Type-2 or CCS2): Charging Sequence

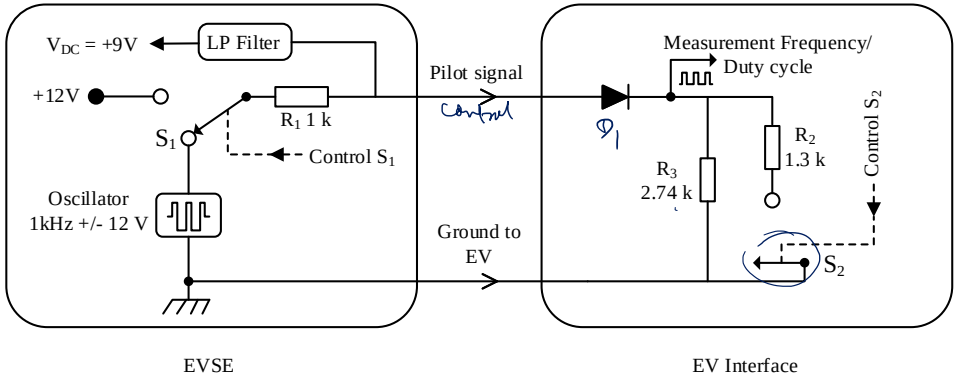


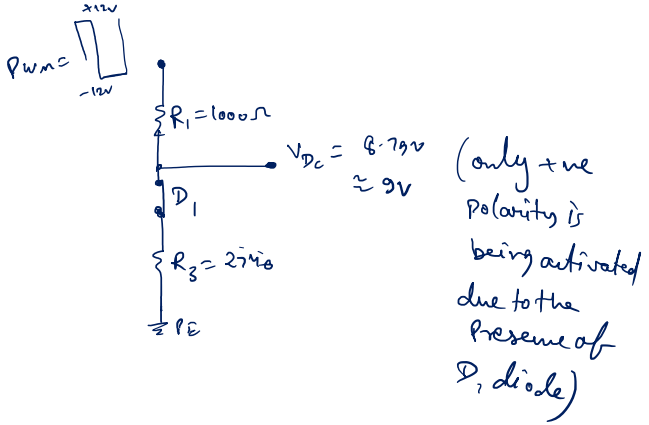


- State-B1
- V_{DC} = +9V
- EV connected but not ready for charging
- Oscillator is still off/PWM is deactivated

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
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AC Charging (Type-2 or CCS2): Charging Sequence



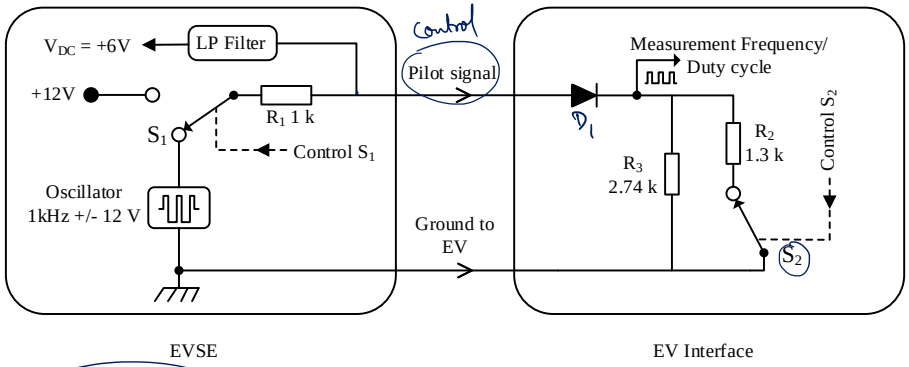


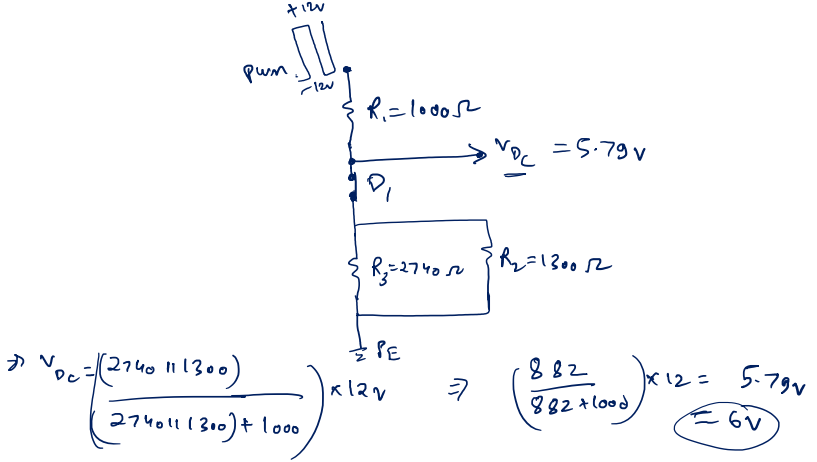
A → B1 → B2 (EVSE is ready)

- State-B2
- V_{DC} = +9V
- EV connected but not ready for charging
- PWM is activated indicates EVSE is ready for energy transfer

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
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AC Charging (Type-2 or CCS2): Charging Sequence





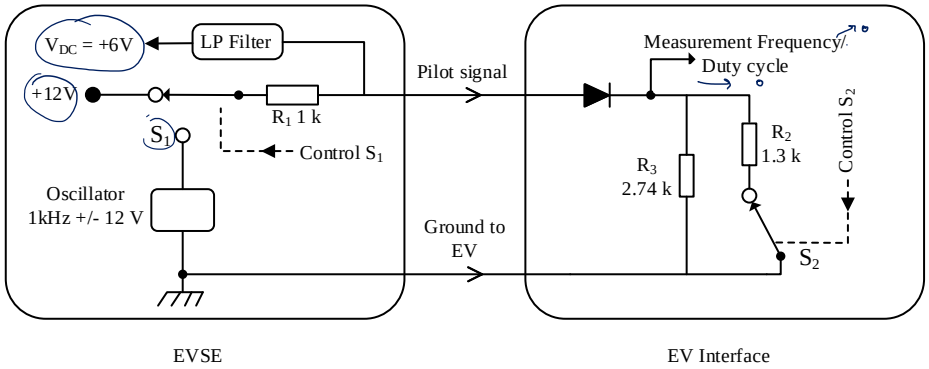
- State- C2
- V_{DC} = +6V
- EV ready for charging by changing state of S_2
- PWM is activated indicates EVSE is ready for energy transfer

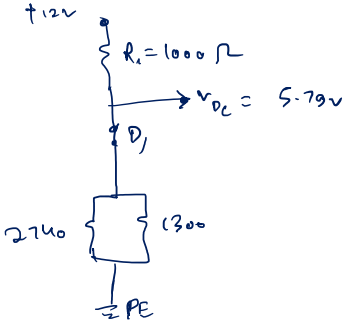
A → B1 → B2 → C2

$2740 \parallel 1300 = \frac{2740 \times 1300}{2740 + 1300} = 882\,\Omega$

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
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AC Charging (Type-2 or CCS2): Charging Sequence



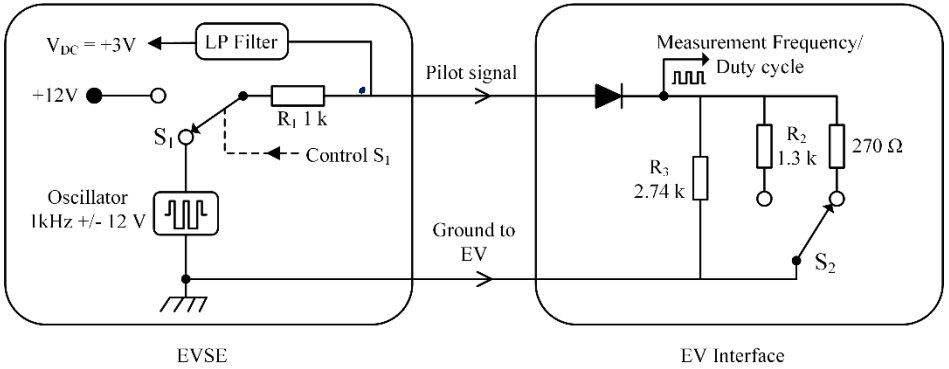


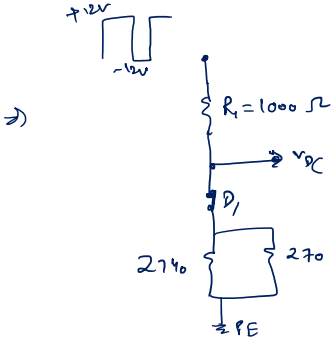
- State- C1
- V_{DC} = +6V
- EV ready for charging
- PWM is de-activated indicates EVSE is not ready

A → B1 → B2 → C2/C1

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
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AC Charging (Type-2 or CCS2): Charging Sequence





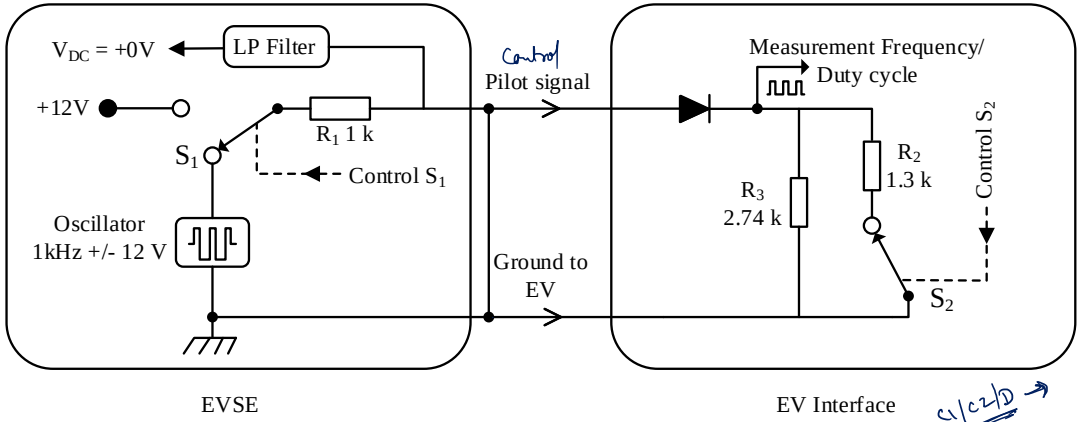
$V_{DC} = \frac{(2740 \parallel 270)}{(2740 \parallel 270) + 1000} \times 12 = \frac{246}{246 + 1000} \times 12 = 2.37\,V \simeq 3\,V$

$2740 \parallel 270 = \frac{2740 \times 270}{2740 \times 270} \simeq 246\,\Omega$

- State- D
- V_{DC} = 3V
- EV ready for charging but requires ventilation
- PWM is activated indicates EVSE is ready for energy transfer

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
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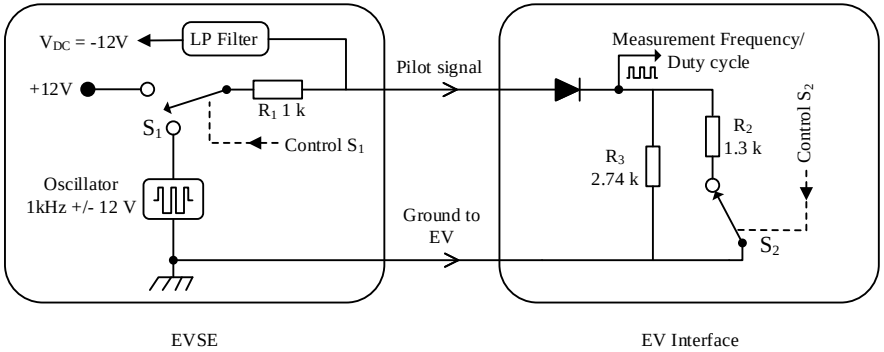
AC Charging (Type-2 or CCS2): Charging Sequence



- State- E
- V_{DC} = 0V
- Problems (like short-circuit)
- EVSE error

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
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AC Charging (Type-2 or CCS2): Charging Sequence



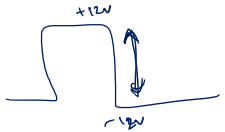
Error condition or EVSE gets plugged out or unavailable

- State- F
- V_{DC} = -12V
- EVSE is not available or error

Courtesy: 1. Rata M,et.al " The ElectricalVehicle Simulator for Charging Station in Mode 3 of IEC 61851-1 Standard. *Energies*. 2020; 13(1):176. https://doi.org/10.3390/en13010176
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Overall Charging Sequence

State-A	State B1	State B2	State C1	State C2	State - D	State - E	State F
$V_{PL} = 12V$	$V_{PL} = 9V$	$V_{PL} = 9V$	$V_{PL} = 6V$	$V_{PL} = 6V$	$V_{PL} = 3V$	$V_{PL} = 0$	$V_{PL} = -12V$
→ EV is not connected to EVSE	→ S_1 is still connected to fixed +12V	S_1 is connected to oscillator	S_1 → fixed 12V	S_1 → oscillator	S_1 → oscillator	fault condition	Error condition
R_{CP-PE} = Not defined	→ PWM is deactivated	→ PWM is activated	S_2 → is on and connected to R_2	S_2 → is on and connected to R_2	S_2 → connected to 270Ω	Communication fault	EVSE is not available
	→ EV is connected	→ EVSE is ready	→ PWM is deactivated	→ PWM is activated	→ EVSE is ready		
+12V	→ Neither EV or EVSE is ready for Energy transfer	→ EV is not ready & S_2 is still not connected to R_2	→ EV is ready	→ EVSE is ready	→ EV is ready	0V	-12V
	R_{CP-PE} = 2740Ω		→ EVSE is not ready	→ EV is already ready	→ But EV requires some kind of ventillation		
	+9V	+9V/-12V	+6V	+6V/-12V	+3V/-12V		
		R_{CP-PE} = 2740Ω	R_{CP-PE} = 882Ω	R_{CP-PE} = 882Ω	R_{CP-PE} = 246Ω		



The details are in given in SAEJ1772 standards (in the form of SAE J1772 signalling protocol) and also adopted in IEC 61851

Duty Ratio of PWM at CP Signal

CP Signal Duty Ratio → is defined by EVSE

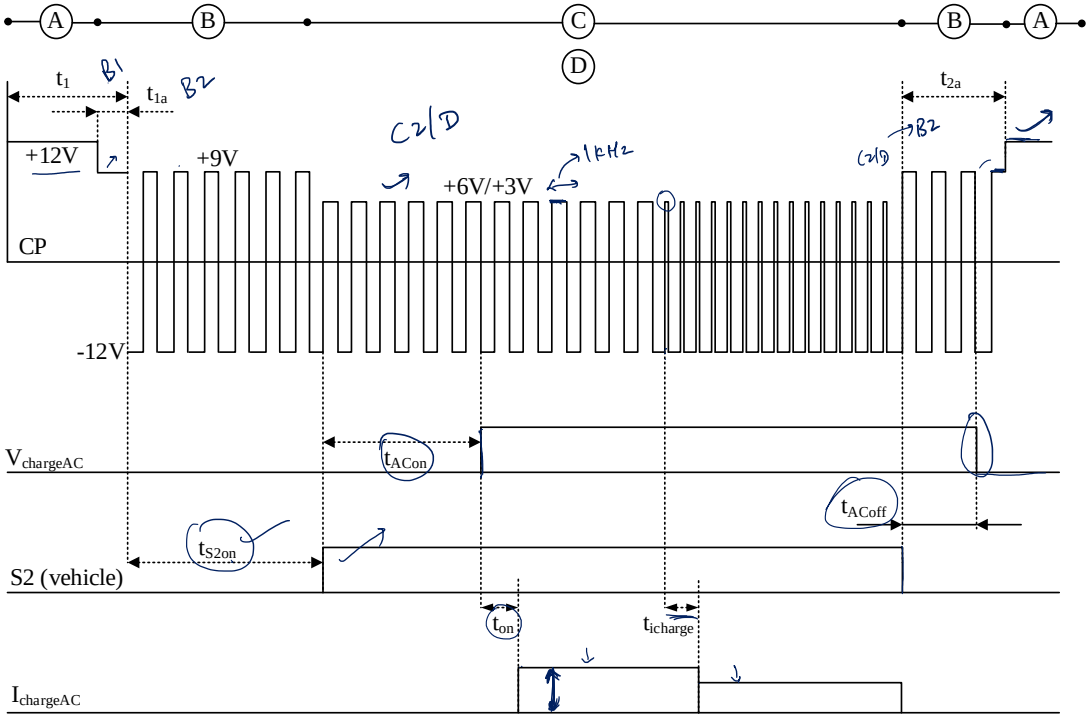
Duty ratio (%)	Maximum current (A) to be drawn by EV
0 ≤ D ≤ 3	Not allowed
3 ≤ D ≤ 7	As indicated by digital communication (high-level communication) for DC charging
7 ≤ D ≤ 8	Not allowed
8 ≤ D ≤ 10	6
10 ≤ D ≤ 85	0.6*D
85 ≤ D ≤ 96	(D - 64) * 2.5
96 ≤ D ≤ 97	80 A
97 ≤ D ≤ 100	Not allowed

$D = 20$
$\Rightarrow I_{max} = 0.6 \times 20$
$= 12\,A$

Maximum current to be drawn by vehicle according to IEC 61851-1

Bahrami, Ali. (2020). EV Charging Definitions, Modes, Levels, Communication Protocols and Applied Standards. 10.13140/RG.2.2.15844.53123/11.

Waveform at CP pin



Source: M. Parchomiuk, A. Moradewicz and H. Gawiński, "An Overview of Electric Vehicles Fast Charging Infrastructure," *2019 Progress in Applied Electrical Engineering (PAEE)*, Koscielisko, Poland, 2019, pp. 1-5, doi: 10.1109/PAEE.2019.8788983

Thank You